

Quiz 1

ECON 320 – Introduction to Mathematical Economics

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Instructions

- Answer all questions. Show all steps clearly to receive full credit.
 - You may use diagrams where appropriate.
 - Total time: 15 minutes.
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Q1. Solve the inequality

$$|2x - 5| < 7$$

and represent the solution set on the real line. Interpret in economic terms if x is quantity and the inequality represents a feasible demand constraint.

Q2. Consider the feasible set defined by

$$S = \{(x, y) \in \mathbb{R}^2 \mid x + y \leq 10, x \geq 0, y \geq 0\}.$$

- Sketch the set S .
- If $y = 2$, what values of x are admissible?
- Interpret this as a resource allocation problem.

Q3. Consider the demand function given implicitly by

$$Q^d(P, Y) = Y - P^2 = 0,$$

where P is price and Y is income.

- Derive $\frac{dP}{dY}$.
- What does the sign of this derivative tell us about the relationship between income and equilibrium price?